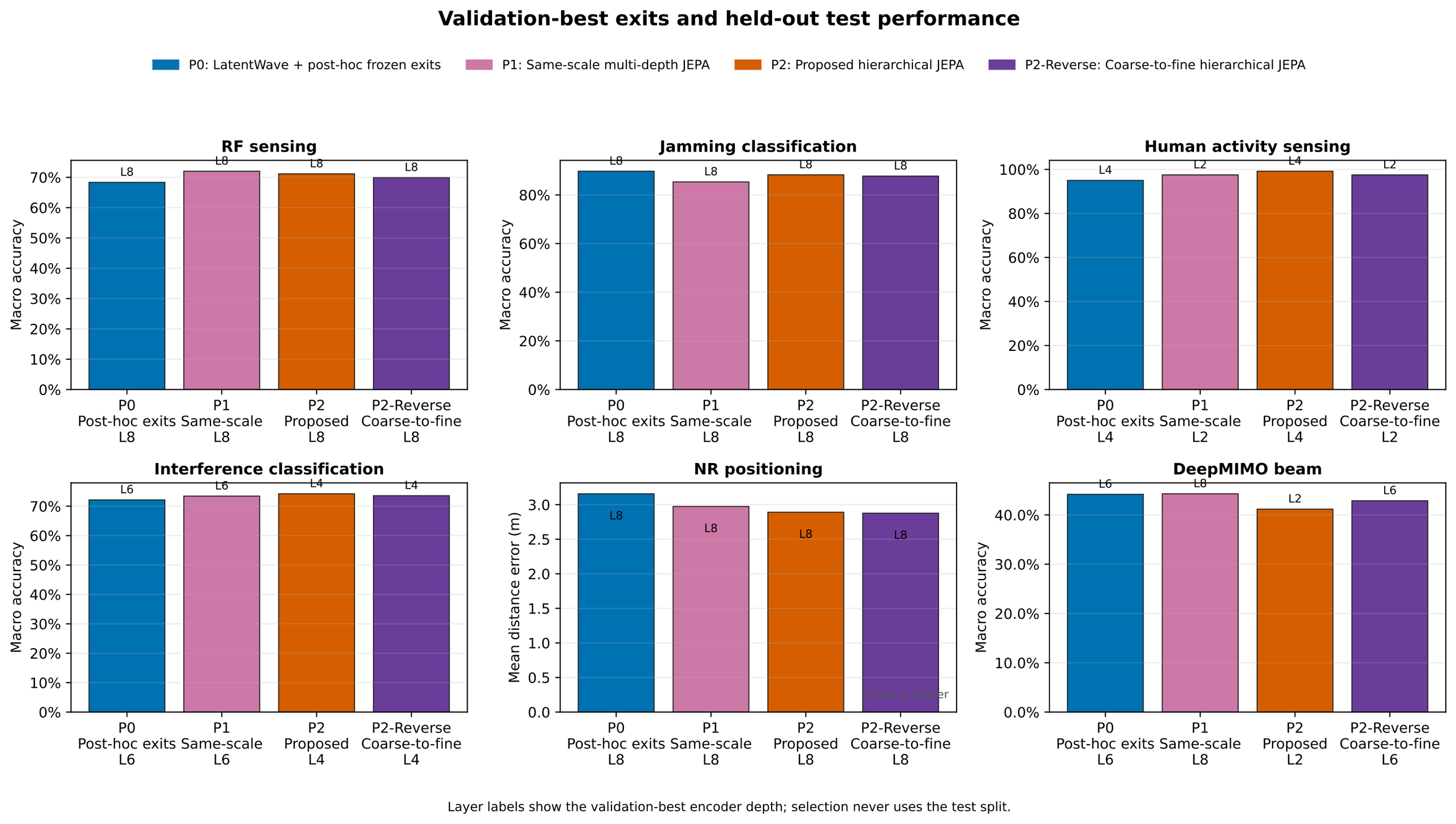
Validation-best exits and held-out test performance
P0: LatentWave + post-hoc frozen exits
P1: Same-scale multi-depth JEPA
P2: Proposed hierarchical JEPA
P2-Reverse: Coarse-to-fine hierarchical JEPA
RF sensing
Macro accuracy
L8
L8
L8
L8
P0
Post-hoc exits
L8
P1
Same-scale
L8
P2
Proposed
L8
P2-Reverse
Coarse-to-fine
L8
Jamming classification
Macro accuracy
L8
L8
L8
L8
P0
Post-hoc exits
L8
P1
Same-scale
L8
P2
Proposed
L8
P2-Reverse
Coarse-to-fine
L8
Human activity sensing
Macro accuracy
L4
L2
L4
L2
P0
Post-hoc exits
L4
P1
Same-scale
L2
P2
Proposed
L4
P2-Reverse
Coarse-to-fine
L2
Interference classification
Macro accuracy
L6
L6
L4
L4
P0
Post-hoc exits
L6
P1
Same-scale
L6
P2
Proposed
L4
P2-Reverse
Coarse-to-fine
L4
NR positioning
Mean distance error (m)
lower is better
L8
L8
L8
L8
P0
Post-hoc exits
L8
P1
Same-scale
L8
P2
Proposed
L8
P2-Reverse
Coarse-to-fine
L8
DeepMIMO beam
Macro accuracy
L6
L8
L2
L6
P0
Post-hoc exits
L6
P1
Same-scale
L8
P2
Proposed
L2
P2-Reverse
Coarse-to-fine
L6
Layer labels show the validation-best encoder depth; selection never uses the test split.